

ASP.NET

Unit-III: ASP.NET Web Forms

Comprehensive Study Notes

BCA Semester V | HPU

3. Introduction to ASP.NET Web Forms

3.1 ASP.NET Web Forms Overview

Definition

ASP.NET Web Forms is a web application framework that allows developers to build dynamic websites using a familiar event-driven programming model. It is part of the .NET Framework and provides a rich set of server controls and automatic state management.

Key Concepts

Concept	Explanation
Web Form	An .aspx page that contains HTML and server controls
Server Controls	Controls that run on the server and generate HTML
Event-Driven	Like Windows Forms, events trigger server-side code
State Management	Automatically maintains control values across postbacks
Code-Behind	Separates presentation (HTML) from logic (VB/C#)

Web Forms Architecture

ASP.NET Web Forms follow a page-based architecture where each page (.aspx) is compiled into a class that inherits from `System.Web.UI.Page`. The architecture includes:

- **Browser** sends an HTTP request for an .aspx page
- **IIS / ASP.NET Runtime** receives the request and creates an instance of the page class
- **Page Life Cycle** executes (Init, Load, Events, Render)
- **HTML** is rendered and sent back as the HTTP response
- **ViewState** automatically persists control state across round trips

3.2 Web Forms Code Models

In-Page Format (Single File)

In the **in-page format**, both HTML markup and server-side code reside in a single .aspx file. Code is enclosed in `<script runat="server">` blocks.

SimpleCalculator.aspx (Single File)

```
<%@ Page Language="VB" AutoEventWireup="true" %>
<!DOCTYPE html>
<html>
<head runat="server">
    <title>Simple Calculator</title>
</head>
<body>
    <form id="form1" runat="server">
        <asp:TextBox ID="txtNum1" runat="server" />
        <asp:TextBox ID="txtNum2" runat="server" />
        <asp:Button ID="btnAdd" runat="server" Text="Add" OnClick="btnAdd_Click" />
        <asp:Label ID="lblResult" runat="server" />
    </form>
    <script runat="server">
        Protected Sub btnAdd_Click(ByVal sender As Object, ByVal e As EventArgs)
            lblResult.Text = "Result: " & (CInt(txtNum1.Text) + CInt(txtNum2.Text))
        End Sub
    </script>
</body>
</html>
```

Code-Behind Format

In the **code-behind format**, presentation (HTML) is separated from logic (VB.NET/C# code). Two files are used: .aspx for markup and .aspx.vb (or .aspx.cs) for code.

Default.aspx (Presentation Layer)

```
<%@ Page Language="VB" AutoEventWireup="false" CodeFile="Default.aspx.vb" Inherits="Default" %>
<!DOCTYPE html>
<html>
<head runat="server">
    <title>Code-Behind Demo</title>
</head>
<body>
    <form id="form1" runat="server">
        <asp:Label ID="lblMessage" runat="server" Text="Hello" />
        <asp:Button ID="btnSubmit" runat="server" Text="Submit" OnClick="btnSubmit_Click" />
    </form>
</body>
</html>
```

Default.aspx.vb (Code-Behind)

```
Partial Class _Default
    Inherits System.Web.UI.Page

    Protected Sub btnSubmit_Click(ByVal sender As Object, ByVal e As EventArgs)
        lblMessage.Text = "Button clicked at " & DateTime.Now.ToString()
    End Sub
End Class
```

3.3 Web Form Life Cycle

The Web Form life cycle consists of multiple events that fire in a specific order during each request:

Event	When It Fires	What You Can Do
PreInit	Before initialization	Set master pages, themes
Init	Controls initialized	Read/initialize control properties
InitComplete	After all controls initialized	Tasks requiring all controls ready
PreLoad	Before Load event	Process postback data
Load	Page loads	Most used - Set properties, bind data
Control Events	Button clicks, etc.	Handle user actions
LoadComplete	After all events	Final data processing
PreRender	Before rendering	Make final changes
Render	HTML generation	Output HTML to response
Unload	Page unloaded	Cleanup resources

Life Cycle Demo - LifeCycle.aspx.vb

```
Partial Class LifeCycle
    Inherits System.Web.UI.Page

    Protected Sub Page_PreInit(ByVal sender As Object, ByVal e As EventArgs) Han
        Response.Write("PreInit<br/>")
    End Sub

    Protected Sub Page_Init(ByVal sender As Object, ByVal e As EventArgs) Handle
        Response.Write("Init<br/>")
    End Sub

    Protected Sub Page_InitComplete(ByVal sender As Object, ByVal e As EventArgs
        Response.Write("InitComplete<br/>")
    End Sub

    Protected Sub Page_PreLoad(ByVal sender As Object, ByVal e As EventArgs) Han
        Response.Write("PreLoad<br/>")
    End Sub

    Protected Sub Page_Load(ByVal sender As Object, ByVal e As EventArgs) Handle
        If Not IsPostBack Then
            Response.Write("Load (First Time)<br/>")
        Else
            Response.Write("Load (PostBack)<br/>")
        End If
    End Sub

    Protected Sub Page_PreRender(ByVal sender As Object, ByVal e As EventArgs) H
        Response.Write("PreRender<br/>")
    End Sub

    Protected Sub Page_Unload(ByVal sender As Object, ByVal e As EventArgs) Hand
        ' Cleanup code here
    End Sub
End Class
```

3.4 Server Controls

Types of Server Controls

Type	Description	Examples
HTML Server Controls	Standard HTML elements with <code>runat="server"</code>	<code><div></code> , <code></code> , <code><input></code> , <code><select></code>
Web Server Controls	Rich .NET controls with automatic HTML rendering	Button, TextBox, Label, DropDownList
Validation Controls	Validate user input	RequiredFieldValidator, RangeValidator
Data Controls	Display and manipulate data	GridView, DetailsView, Repeater
Navigation Controls	Site navigation	Menu, TreeView, SiteMapPath
Login Controls	Authentication and membership	Login, CreateUserWizard, PasswordRecovery

Common Properties of Server Controls

Property	Type	Description
ID	String	Unique identifier to access the control in code
runat	String	Must be "server" for server-side processing
Text	String	Display text for the control
Visible	Boolean	Whether the control is rendered on the page
Enabled	Boolean	Whether the control is interactive

CssClass	String	CSS class for styling the control
BackColor	Color	Background color
ForeColor	Color	Foreground (text) color
Font	FontInfo	Font properties (Name, Size, Bold, etc.)
Width / Height	Unit	Dimensions of the control
ToolTip	String	Tooltip text shown on hover
TabIndex	Integer	Tab order index

Standard Controls Reference

Control	Purpose	Key Properties
Label	Display text on the page	Text, Font, ForeColor
TextBox	Accept user text input	Text, TextMode, AutoPostBack, MaxLength
Button	Trigger an action	Text, OnClick, OnClientClick
LinkButton	Button rendered as a hyperlink	Text, OnClick
ImageButton	Button with an image	ImageUrl, OnClick, AlternateText
HyperLink	Navigate to a URL	Text, NavigateUrl, ImageUrl, Target
DropDownList	Single-selection drop-down	Items, SelectedValue, AutoPostBack
ListBox	List with single/multiple selection	Items, SelectionMode, Rows
CheckBox	Boolean on/off selection	Checked, Text, AutoPostBack

RadioButton	Single choice among options	Checked, Text, GroupName
Panel	Container for grouping controls	Visible, BackColor, BorderStyle
Placeholder	Empty container for dynamic controls	Visible
Image	Display an image	ImageUrl, AlternateText, Width, Height
Calendar	Date picker control	SelectedDate, VisibleDate, SelectionMode
FileUpload	Upload files from client	HasFile, FileName, SaveAs()

3.5 Event Handling

Events in Web Forms work on a **PostBack** mechanism. When a user interacts with a control:

1. Control triggers a client-side JavaScript event
2. JavaScript submits the form to the server (PostBack)
3. Server identifies which control triggered the event
4. Server executes the corresponding event handler
5. Server sends updated HTML back to the browser

Common Page and Control Events

Event	Control	When Fires
Page_Load	Page	Every time the page loads (check IsPostBack)
Click	Button, LinkButton, ImageButton	When the button is clicked
TextChanged	TextBox	When text changes (requires AutoPostBack)
SelectedIndexChanged	DropDownList, ListBox	When selection changes (requires AutoPostBack)
CheckedChanged	CheckBox, RadioButton	When checked state changes (requires AutoPostBack)

Button Click Event

```
<%@ Page Language="VB" AutoEventWireup="false" CodeFile="ButtonDemo.aspx.vb" Inherits="ButtonDemo" %>
<!DOCTYPE html>
<html>
<head runat="server"><title>Button Demo</title></head>
<body>
    <form id="form1" runat="server">
        <asp:TextBox ID="txtName" runat="server" />
        <asp:Button ID="btnShow" runat="server" Text="Show Message" OnClick="btnShow_Click" />
        <asp:Label ID="lblMessage" runat="server" />
    </form>
</body>
</html>
```

```
' ButtonDemo.aspx.vb
Partial Class ButtonDemo
    Inherits System.Web.UI.Page

    Protected Sub btnShow_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnShow.Click
        lblMessage.Text = "Hello, " & txtName.Text & "!"
    End Sub
End Class
```

TextBox Events with AutoPostBack

```
' TextBoxDemo.aspx.vb
Partial Class TextBoxDemo
    Inherits System.Web.UI.Page

    Protected Sub txtInput_TextChanged(ByVal sender As Object, ByVal e As EventArgs) Handles txtInput.TextChanged
        lblOutput.Text = "You typed: " & txtInput.Text
    End Sub
End Class
```

IsPostBack Property

The **IsPostBack** property indicates whether the page is being loaded for the first time (false) or in response to a client postback (true). It is used in `Page_Load` to avoid re-initializing data on every postback.

```
Protected Sub Page_Load(ByVal sender As Object, ByVal e As EventArgs) Handles Me
    If Not IsPostBack Then
        ' First time loading – bind controls, set defaults
        ddlCountry.DataSource = GetCountries()
        ddlCountry.DataBind()
    End If
End Sub
```

AutoPostBack Property

AutoPostBack is a property that automatically sends the page back to the server when a control's state changes. Controls like `TextBox`, `CheckBox`, `RadioButton`, `DropDownList`, and `ListBox` support this property.

```
<asp:DropDownList ID="ddlCountry" runat="server" AutoPostBack="true" OnSelectedI
    <asp:ListItem Value="IN">India</asp:ListItem>
    <asp:ListItem Value="US">USA</asp:ListItem>
</asp:DropDownList>
```

3.6 State Management

ASP.NET Web Forms provide several techniques for maintaining state across HTTP requests (which are stateless by nature):

Technique	Scope	Duration	Location	Use Case
ViewState	Page-level	Single page life	Hidden field in page	Preserving control values across postbacks
Session	User-level	Session timeout or browser close	Server memory	Storing user-specific data (cart, login)
Application	Application-level	Application lifetime	Server memory	Global data accessible to all users
Cache	Application-level	Configurable expiry	Server memory	Caching expensive data with expiration
Cookie	User-level	Configurable expiry	Client browser	Persistent user preferences
QueryString	Page-level	Single request	URL	Passing values between pages
HiddenField	Page-level	Single page life	Hidden field in page	Storing small page-specific values

ViewState Demo

```
' ViewStateDemo.aspx.vb
Partial Class ViewStateDemo
    Inherits System.Web.UI.Page

    Protected Sub btnIncrement_Click(ByVal sender As Object, ByVal e As EventArgs)
        Dim count As Integer = CInt(ViewState("Counter"))
        count += 1
        ViewState("Counter") = count
        lblCounter.Text = "Count: " & count.ToString()
    End Sub
End Class
```

Session State Demo

```
' SessionDemo.aspx.vb
Partial Class SessionDemo
    Inherits System.Web.UI.Page

    Protected Sub Page_Load(ByVal sender As Object, ByVal e As EventArgs) Handle
        If Not IsPostBack Then
            Session("UserName") = "John"
        End If
    End Sub

    Protected Sub btnShow_Click(ByVal sender As Object, ByVal e As EventArgs) Ha
        lblSession.Text = "Welcome, " & Session("UserName").ToString()
    End Sub
End Class
```

Application State

```
' Global.asax – Application level
Sub Application_Start(ByVal sender As Object, ByVal e As EventArgs)
    Application("TotalVisits") = 0
End Sub

Sub Session_Start(ByVal sender As Object, ByVal e As EventArgs)
    Application.Lock()
    Application("TotalVisits") = CInt(Application("TotalVisits")) + 1
    Application.Unlock()
End Sub
```

Cookie Example

```
' Writing a cookie
Dim userCookie As New HttpCookie("Preferences")
userCookie("Theme") = "Dark"
userCookie("Language") = "en-US"
userCookie.Expires = DateTime.Now.AddDays(30)
Response.Cookies.Add(userCookie)

' Reading a cookie
If Request.Cookies("Preferences") IsNot Nothing Then
    Dim theme As String = Request.Cookies("Preferences")("Theme")
End If
```


QueryString Example

```
' Page1.aspx – pass data
Response.Redirect("Page2.aspx?id=101&name=John")

' Page2.aspx.vb – receive data
Protected Sub Page_Load(ByVal sender As Object, ByVal e As EventArgs) Handles Me
    If Request.QueryString("id") IsNot Nothing Then
        Dim id As String = Request.QueryString("id")
        Dim name As String = Request.QueryString("name")
    End If
End Sub
```

HiddenField Example

```
<asp:HiddenField ID="hfUserId" runat="server" Value="0" />

' In code-behind
hfUserId.Value = "42"
Dim userId As String = hfUserId.Value
```

3.7 HTML Server Controls

HTML server controls are standard HTML elements with the `runat="server"` attribute that makes them accessible in server-side code. They provide a way to programmatically control HTML elements on the server.

HTML Element	Server Control Class	Key Properties
<div>	HtmlGenericControl	InnerText, InnerHtml, Style
	HtmlGenericControl	InnerText, InnerHtml
<input type="text">	HtmlInputText	Value, Size, MaxLength
<input type="submit">	HtmlInputButton	Value, OnServerClick
<select>	HtmlSelect	Value, Items, SelectedIndex
<textarea>	HtmlTextArea	Value, Rows, Cols
<table>	HtmlTable	Rows, Cells, Border
	HtmlImage	Src, Alt, Width, Height
<a>	HtmlAnchor	HRef, InnerText, Target
<form>	HtmlForm	Action, Method

HTML Controls Example

```
<%@ Page Language="VB" AutoEventWireup="false" CodeFile="HTMLControls.aspx.vb" I
<!DOCTYPE html>
<html>
<head runat="server"><title>HTML Controls</title></head>
<body>
    <form id="form1" runat="server">
        <input id="txtName" type="text" runat="server" />
        <input id="btnSubmit" type="submit" runat="server" value="Submit" onserv
        <span id="lblMessage" runat="server"></span>
    </form>
</body>
</html>
```

```
' HTMLControls.aspx.vb
Partial Class HTMLControls
    Inherits System.Web.UI.Page

    Protected Sub btnSubmit_ServerClick(ByVal sender As Object, ByVal e As Event
        lblMessage.InnerHtml = "Hello, " & txtName.Value
    End Sub
End Class
```

3.8 Web Server Controls

Web server controls are **.NET controls** that provide rich functionality and generate HTML automatically. They include auto-postback, themes, templates, and complex behaviors.

```
<%@ Page Language="VB" AutoEventWireup="false" CodeFile="WebControls.aspx.vb" In
<!DOCTYPE html>
<html>
<head runat="server"><title>Web Controls Demo</title></head>
<body>
    <form id="form1" runat="server">
        <asp:Label ID="lblName" runat="server" Text="Enter Name:" />
        <asp:TextBox ID="txtName" runat="server" />
        <br/>
        <asp:Button ID="btnSubmit" runat="server" Text="Submit" OnClick="btnSubm
        <asp:LinkButton ID="lnkSubmit" runat="server" Text="Or Click Here" OnCli
        <br/>
        <asp:CheckBox ID="chkSubscribe" runat="server" Text="Subscribe" AutoPost
        <br/>
        <asp:RadioButton ID="rbMale" runat="server" GroupName="Gender" Text="Mal
        <asp:RadioButton ID="rbFemale" runat="server" GroupName="Gender" Text="F
        <br/>
        <asp:DropDownList ID="ddlCity" runat="server" AutoPostBack="true" OnSele
            <asp:ListItem Text="Select City" Value="" />
            <asp:ListItem Text="Delhi" Value="DL" />
            <asp:ListItem Text="Mumbai" Value="MH" />
        </asp:DropDownList>
    </form>
</body>
</html>
```


3.9 Complete Example: Student Registration Form

```
<%@ Page Language="VB" AutoEventWireup="false" CodeFile="StudentRegister.aspx.vb"
<!DOCTYPE html>
<html>
<head runat="server"><title>Student Registration</title></head>
<body>
    <form id="form1" runat="server">
        <h3>Student Registration Form</h3>

        <asp:Label ID="lblName" runat="server" Text="Full Name:" />
        <asp:TextBox ID="txtName" runat="server" /><br/>

        <asp:Label ID="lblEmail" runat="server" Text="Email:" />
        <asp:TextBox ID="txtEmail" runat="server" /><br/>

        <asp:Label ID="lblGender" runat="server" Text="Gender:" />
        <asp:RadioButton ID="rbMale" runat="server" GroupName="Gender" Text="Mal
        <asp:RadioButton ID="rbFemale" runat="server" GroupName="Gender" Text="F

        <asp:Label ID="lblCourse" runat="server" Text="Course:" />
        <asp:DropDownList ID="ddlCourse" runat="server">
            <asp:ListItem Text="Select" Value="" />
            <asp:ListItem Text="BCA" Value="BCA" />
            <asp:ListItem Text="MCA" Value="MCA" />
            <asp:ListItem Text="B.Tech" Value="BTech" />
        </asp:DropDownList><br/>

        <asp:Label ID="lblHobbies" runat="server" Text="Hobbies:" />
        <asp:CheckBox ID="chkReading" runat="server" Text="Reading" />
        <asp:CheckBox ID="chkMusic" runat="server" Text="Music" />
        <asp:CheckBox ID="chkSports" runat="server" Text="Sports" /><br/>

        <asp:Button ID="btnRegister" runat="server" Text="Register" OnClick="btn
        <asp:Button ID="btnReset" runat="server" Text="Reset" OnClick="btnReset_
        <hr/>
        <asp:Label ID="lblResult" runat="server" ForeColor="#1a5276" />
    </form>
</body>
</html>
```

```
' StudentRegister.aspx.vb
Partial Class StudentRegister
    Inherits System.Web.UI.Page

    Protected Sub btnRegister_Click(ByVal sender As Object, ByVal e As EventArgs)
        Dim gender As String = "Not selected"
        If rbMale.Checked Then gender = "Male"
        If rbFemale.Checked Then gender = "Female"

        Dim hobbies As String = ""
        If chkReading.Checked Then hobbies &= "Reading "
        If chkMusic.Checked Then hobbies &= "Music "
        If chkSports.Checked Then hobbies &= "Sports"

        lblResult.Text = "<b>Registration Successful!</b><br/>" &
            "Name: " & txtName.Text & "<br/>" &
            "Email: " & txtEmail.Text & "<br/>" &
            "Gender: " & gender & "<br/>" &
            "Course: " & ddlCourse.SelectedValue & "<br/>" &
            "Hobbies: " & hobbies
    End Sub

    Protected Sub btnReset_Click(ByVal sender As Object, ByVal e As EventArgs) H
        txtName.Text = ""
        txtEmail.Text = ""
        rbMale.Checked = False
        rbFemale.Checked = False
        ddlCourse.SelectedIndex = 0
        chkReading.Checked = False
        chkMusic.Checked = False
        chkSports.Checked = False
        lblResult.Text = ""
    End Sub
End Class
```

Quick Reference Summary

Common Web Server Controls Reference

Control	Tag	Purpose
Label	<asp:Label>	Display read-only text
TextBox	<asp:TextBox>	Accept user input
Button	<asp:Button>	Submit form / trigger action
LinkButton	<asp:LinkButton>	Button styled as a link
ImageButton	<asp:ImageButton>	Button with an image
HyperLink	<asp:HyperLink>	Navigate to a URL
DropDownList	<asp:DropDownList>	Single-selection drop-down
ListBox	<asp:ListBox>	List with single/multiple selection
CheckBox	<asp:CheckBox>	Boolean selection
RadioButton	<asp:RadioButton>	Single choice among options
Panel	<asp:Panel>	Container for grouping
Placeholder	<asp:Placeholder>	Empty container for dynamic controls
Image	<asp:Image>	Display an image
Calendar	<asp:Calendar>	Date picker
FileUpload	<asp:FileUpload>	Upload files

Common Events

Event	Control	When Fires
Click	Button, LinkButton, ImageButton	When clicked
TextChanged	TextBox	When text changes (AutoPostBack)
SelectedIndexChanged	DropDownList, ListBox	When selection changes
CheckedChanged	CheckBox, RadioButton	When checked state changes

State Management Comparison

Feature	ViewState	Session	Application	Cookie	QueryString
Location	Client (hidden field)	Server memory	Server memory	Client browser	URL
Data size	Small	Medium	Medium	Small	Very small
Security	Low (encoded)	High	High	Low	Low (visible)
Lifetime	Single page	Session timeout	App restart	Expiry set	Single request
Scope	Single page	Single user	All users	Single user	Target page

Questions

Q1. What is ASP.NET Web Forms? Explain its key concepts and architecture.

Q2. Differentiate between In-Page Format and Code-Behind Format in ASP.NET Web Forms with suitable examples.

Q3. Explain the Web Form Life Cycle in detail. List all the events in the correct order.

Q4. What are Server Controls in ASP.NET? List and explain the different types of server controls.

Q5. Explain the common properties of server controls: ID, runat, Text, Visible, Enabled, and CssClass.

Q6. Write short notes on any five standard controls in ASP.NET Web Forms.

Q7. Explain event handling in ASP.NET Web Forms. Describe Click, TextChanged, SelectedIndexChanged, and CheckedChanged events with examples.

Q8. What is the use of IsPostBack property? Explain with a code example.

Q9. What is AutoPostBack? Explain with an example.

Q10. Explain the various state management techniques available in ASP.NET Web Forms.

Q11. Differentiate between ViewState, Session, and Application state with suitable examples.

Q12. How are cookies and QueryString used for state management? Explain with examples.

Q13. Explain HTML Server Controls. How are they different from Web Server Controls?**

Q14. Write a complete ASP.NET program for a Student Registration Form using Web Server Controls.

Q15. Explain the Page_Load event and its importance in ASP.NET Web Forms.